

Sub-Circuit	-	Circuit name	-	Component	-	Value/Type	-	Reference	-	Pins	-	NOTES
		IR Signal		IR Receiver		Digital 38KHz IR Sensor Module for Arduino		IRS		1,2,3		pin 1 (GND) pin 2 (VCC) pin 3 (DATA)
T2 control signal	-	Lights Out	-	Resistor	-	100K ohms	-	R3	-	1,2	-	
EM Switch Coil	-	Lights Out	-	Transistor	-	2N2222	-	T2	-	1,2,3	-	pin 1 (emitter) pin 2 (base) pin 3 (collector)
	-	12 to 6V Drop Down	-	Resistor	-	100K ohms	-	R4	-	1,2	-	
	-	12 to 6V Drop Down	-	Resistor	-	100K ohms	-	R5	-	1,2	-	
	-	+Vtest	-	Resistor	-	100K ohms	-	R1	-	1,2	-	
	-	+Vtest	-	Resistor	-	100K ohms	-	R2	-	1,2	-	
+Vtest Voltage Monitor	-	Voltage Comparitor	-	Low Power Dual Operational Amplifier	-	LM358	-	U1	-	1,2,3,4,5,6,7,8	-	pin 8 (VCC) pin 4 (GND) pin 3 (+V TEST) pin 2 (-V REF) pin 1 (OUTPUT1)
	-	-Vref	-	Resistor	-	100K ohms	-	R6	-	1,2	-	
	-	-Vref	-	Resistor	-	100K ohms	-	R7	-	1,2	-	
	-	-Vref	-	Resistor	-	100K ohms	-	R8	-	1,2	-	
Output 1 Indicator	-	Output 1 Indicator	-	Led	-	Green	-	L1	-	1,2	-	pin 1 (+ terminal) pin 2 (- terminal)
	-	Output 1 Indicator	-	Resistor	-	100K ohms	-	R9	-	1,2	-	
T1 control signal	-	EM Switch	-	Resistor	-	100K ohms	-	R10	-	1,2	-	
	-	EM Switch	-	Transistor	-	2N2222	-	T1	-	1,2,3	-	pin 1 (emitter) pin 2 (base) pin 3 (collector)
EM Switch	-	EM Switch	-	Diode	-	1N4001	-	D1	-	1,2	-	pin 1 (anode) pin 2 (cathode)
	-	EM Switch	-	EM Switch	-	Nilight 50003R Automotive Set 5-Pin 30/40A 12V SPDT	-	EMS1	-	30,85,86,87,87a	-	30 (High Power Output) 85 (Relay Coil Ground) 86 (Relay Coil Feed - Trigger Wire) 87 (High Power Output - Normally Open Contact) 87a (High Power Output - Normally Closed Contact)
T3 control signal	-	Light On	-	Resistor	-	100K ohms	-	R11	-	1,2	-	
	-	Light On	-	Transistor	-	2N2222	-	T3	-	1,2,3	-	pin 1 (emitter) pin 2 (base) pin 3 (collector)
	-		-	Controller	-	Arduino Uno	-	A1	-	2,3,4,5,GND	-	pin 2 (IRS Signal) pin 3 (T2 On/Off) pin 4 (T3 On/Off) pin 5 (Output) GND (must be connected to the circuit board GND to avoid a floating ground)